DESCENT OF THE COMPLETELY DECOMPOSED PROPERTY ALONG A LIMIT OF SCHEMES WITH AFFINE TRANSITIONS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 23257-23290, immediately following the statement at 23239-23255.

Editorial note. Four cross-chapter pointers (Limits) plus a same-chapter lemma are recorded in assembly.external_references.

Editorial note. The statement fixes the indexing convention (0 in Lambda, base change to S_lambda); that wording is retained verbatim in the statement span.

Allowed standard prerequisites (editorial).

- Decomposition of completely decomposed morphisms. chapter lemma lemma-decompose, cited in the proof. Provides the decomposition criterion that the descended data has to satisfy.
- Local finite presentation along a limit. Limits, proposition limits-proposition-characterize-locally-finite-presentation, cited in the proof. Certifies local finite presentation from the presentation at a stage.
- Descending finite presentation, immersions and surjectivity. Limits, lemmas limits-lemma-descend-finite-presentation, limits-lemma-descend-immersion and limits-lemma-descend-surjective, cited in the proof. Move the finitely presented pieces, the immersions and the surjection from X to some stage X_lambda.

Source reference. Original lemma 0GTL: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 23187–23187. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01ZM: focused-seeds/geometry/S9-morph-descend-cd/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0GTB: focused-seeds/geometry/S9-morph-descend-cd/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07RR: focused-seeds/geometry/S9-morph-descend-cd/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 01ZC: focused-seeds/geometry/S9-morph-descend-cd/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $S = \lim_{\lambda \in \Lambda} S_\lambda$ be a limit of a directed system of schemes with affine transition morphisms. Let $0 \in \Lambda$ and let $f_0 : X_0 \to Y_0$ be a morphism of schemes over S_0. For $\lambda \geq 0$ let $f_\lambda : X_\lambda \to Y_\lambda$ be the base change of f_0 to S_λ and let $f : X \to Y$ be the base change of f_0 to S. If*

(1) *f is completely decomposed,*
(2) *Y_0 is quasi-compact and quasi-separated, and*
(3) *f_0 is locally of finite presentation,*

then there exists an $\lambda \geq 0$ such that f_λ is completely decomposed.

Proof. Since Y_0 is quasi-compact and quasi-separated, the scheme Y, which is affine over Y_0, is quasi-compact and quasi-separated. Choose $n \geq 0$ and $Z_i \to Y$, $i = 1, \ldots, n$ as in Lemma 0GTL. Denote $\sigma_i : Z_i \to X$ morphisms over Y which exist by our choice of Z_i. After increasing $0 \in \Lambda$ we may assume there exist morphisms $Z_{i,0} \to Y_0$ of finite presentation whose base changes to S are the morphisms $Z_i \to Y$, see Limits, Lemma 01ZM. By Limits, Lemma 0GTB we may assume, after possibly increasing 0, that $Z_{i,0} \to Y_0$ is an immersion. Since $\coprod Z_i \to Y$ is surjective, we may assume, after possibly increasing 0, that $\coprod Z_{i,0} \to Y_0$ is surjective, see Limits, Lemma 07RR. Observe that $Z_i = \lim_{\lambda \geq 0} Z_{i,\lambda}$ where $Z_{i,\lambda} = Y_\lambda \times_{Y_0} Z_{i,0}$. Let us view the compositions

$$Z_i \xrightarrow{\sigma_i} X \to X_0$$

as morphisms over Y_0. Since f_0 is locally of finite presentation by Limits, Proposition 01ZC we can find a $\lambda \geq 0$ such that there exist morphisms $\sigma'_{i,\lambda} : Z_{i,\lambda} \to X_0$ over Y_0 whose precomposition with $Z_i \to Z_{i,\lambda}$ are the displayed arrows. Of course, then $\sigma'_{i,\lambda}$ determines a morphism $\sigma_{i,\lambda} : Z_{i,\lambda} \to X_\lambda = X_0 \times_{Y_0} Y_\lambda$ over Y_λ. Since $\coprod Z_{i,\lambda} \to Y_\lambda$ is surjective we conclude that $X_\lambda \to Y_\lambda$ is completely decomposed. □